

Carhart (1997) added a fourth factor representing cross-sectional momentum.

Academics then became factor happy. At least 82 of them have been published in leading academic journals. Searching diligently for explanatory factors is reminiscent of Procrustes, the Greek mythological figure who made his visitors fit his bed by either stretching them or cutting off their legs.

Data snooping biases can arise from focusing on anomalous data. Harvey, Liu, and Zhu (2014), after applying to a variety of market risk factors a significance deflation adjustment for data mining bias, concluded, "... many of the factors discovered in the field of finance are likely false discoveries Echoing a recent disturbing conclusion in the medical literature, we argue that most claimed research findings in financial economics are likely false."⁷

CAPM was just not that reliable. For followers of CAPM, the real world was an annoying special case. Empirical tests showed that high beta and high volatility stocks did not give the return advantage that they should, according to three- or four-factor CAPM. Fama and French (2004) stepped forward and called CAPM "empirically vacuous." They noted "whether the model's problems reflect weakness in the theory or in its empirical implementation, the failure of the CAPM in empirical tests implies that most applications of the model are invalid."

CAPM not only had empirical problems. It, along with other models of modern finance, also had theoretical issues. Financial models generally rely on two main assumptions. The first is that market prices adhere to a normal or lognormal distribution.⁸ The second assumption of modern finance is that prices are independent of one another. Yesterday's prices should have no influence on today's prices.

Mandelbrot (2004) courageously challenged both these assumptions. He demonstrated that market prices do not follow a normal distribution, but rather one associated with unstable variance and fatter tails denoting a higher frequency of extreme events. Mandelbrot identified this as a Cauchy distribution. Others have called it a stable Paretian, Pareto-Levy, or Levy-Mandelbrot distribution. Whatever you call it, this distribution means that catastrophic drops in stock prices can happen more frequently than a normal distribution would suggest.